Matrix and Tensor Factorizations for Cross-Domain Recommendations in Books and Products

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***Abstract*—Recommender systems enhance the user experience by providing personalized suggestions based on user preferences and historical behavior. This report investigates the development and evaluation of two recommendation systems: one for predicting product ratings and another for recommending books. Using real-world user reviews and rating datasets, we preprocess the data to construct user-item matrices and apply Singular Value Decomposition (SVD) to uncover latent factors. The models are evaluated based on their predictive performance using reconstructed ratings, focusing on personalization accuracy. Additionally, the report outlines a methodology for selecting and visualizing top recommendations for individual users, enabling actionable insights into user preferences. The visualization techniques employed include bar plots of predicted top-rated items, ensuring an intuitive representation of recommendations. This work highlights the efficacy of SVD in generating meaningful recommendations, addresses challenges associated with sparse data, and underscores the importance of visualization in user-centric recommendation systems. Future work aims to incorporate advanced techniques like hybrid models and contextual awareness for improved accuracy and adaptability.**

I. Introduction and Statement of the Problem

In today's digital era, the exponential growth of online platforms has transformed the way users interact with products and services. Recommender systems have become indispensable tools for businesses, enabling them to personalize user experiences by analyzing behavioral data. From suggesting books to recommending products, these systems play a pivotal role in retaining user engagement, improving satisfaction, and driving revenue. Recommender systems achieve this by predicting user preferences and generating a ranked list of items tailored to individual needs. Despite their wide adoption, building an efficient recommendation system remains a complex challenge, primarily due to the high dimensionality and sparsity of user-item data. Users often interact with only a small fraction of available items, resulting in sparse datasets that hinder traditional predictive methods. Additionally, the diversity of user preferences and the dynamic nature of interactions necessitate robust techniques capable of uncovering latent patterns in the data. This project explores the application of Singular Value Decomposition (SVD), a matrix factorization technique, to address the challenges of sparsity and high dimensionality in recommender systems. The study includes product reviews and book review datasets, which provide user-generated ratings, timestamps, and textual summaries. The project investigates the effectiveness of SVD in generating personalized recommendations by reconstructing user-item matrices and identifying latent factors influencing user preferences. The significance of this work lies in its dual focus: analyzing recommendation performance for distinct domains (products and books) and visualizing the top recommendations for individual users. The incorporation of data visualization not only enhances interpretability but also provides actionable insights for system designers and business stakeholders. By sampling and preprocessing large datasets, the project aims to strike a balance between computational efficiency and recommendation accuracy. User-item interaction data is often sparse, with many missing values, which complicates the task of predicting preferences. The study explores how SVD can effectively reduce dimensionality and capture latent user-item interactions. While generic recommendation methods suggest popular items, they often fail to provide personalized suggestions that align with individual user preferences. The project aims to bridge this gap using matrix factorization techniques. Different domains, such as product reviews and book reviews, exhibit varying user behaviors and interaction patterns. The study seeks to evaluate and compare recommendation systems across these domains to understand their unique challenges and opportunities. Recommendations must not only be accurate but also interpretable for users and stakeholders. The project addresses this by visualizing top-rated items for users, providing clarity on why certain

recommendations are made.

II. Review of Literature

Cross-domain recommendation systems have attracted significant attention in recent years due to their ability to improve recommendation accuracy and overcome challenges such as data sparsity and the cold-start problem. These systems leverage knowledge from multiple domains to enhance recommendation quality when data from a single domain is insufficient. Pourheidari (2019) investigated the use of matrix and tensor factorization techniques for cross-domain recommendation systems. The study emphasizes the role of tensor factorizations in handling multi-relational data and extracting hidden patterns across different domains. The research demonstrated that tensor-based methods are particularly effective in dealing with sparse datasets, and the approach outperforms traditional matrix factorization models in terms of recommendation accuracy. By exploring cross-domain relationships, Pourheidari (2019) concluded that tensor factorization could significantly improve recommendation systems where user-item interactions are limited, which is a common challenge in many real-world scenarios [1]. Taneja and Arora (2018) proposed a multidimensional tensor factorization framework that aims to utilize auxiliary data from multiple domains to improve recommendation performance. Their method incorporates multiple types of data, such as user-item interactions and domain-specific metadata, to create a more comprehensive recommendation model. They observed that their multidimensional tensor factorization approach resulted in higher recommendation accuracy compared to traditional collaborative filtering methods, particularly in cases where the overlap between domains was minimal. The study demonstrated that leveraging data from multiple domains could mitigate the impact of sparsity and enhance personalization in cross-domain recommendation systems [2]. Cantador et al. (2015) offered a comprehensive review of cross-domain recommender systems, providing an in-depth analysis of various techniques, including matrix factorization, neighborhood-based methods, and content-based approaches. Their work highlights the importance of semantic matching and latent factor models in addressing the challenges posed by domain heterogeneity. Cantador et al. (2015) emphasized the potential of hybrid methods that combine collaborative filtering with domain-specific content to enhance recommendation accuracy. Their review provides a solid foundation for understanding how different cross-domain recommendation strategies can be applied to real-world problems, including e-commerce, social media, and entertainment platforms [3]. Song et al. (2017) introduced a joint tensor factorization approach to cross-domain recommendation, which aims to exploit inter-domain correlations more effectively. The proposed method combines the advantages of both matrix factorization and tensor decomposition, allowing the system to capture complex relationships between users and items across different domains. Through empirical evaluations, the authors demonstrated that their approach significantly improved recommendation performance in scenarios with limited overlap between domains. Their study also highlighted the flexibility of joint tensor factorization in accommodating multiple domains with diverse characteristics, making it suitable for applications where data from multiple sources must be integrated [4]. Kumar et al. (2015) explored the integration of semantic similarity measures with tensor decomposition to enhance cross-domain recommendation accuracy. By incorporating external knowledge sources such as ontologies and semantic embeddings, their method improves the interpretability and effectiveness of recommendations. The authors found that semantic similarity between items from different domains could be captured effectively through tensor decomposition, leading to more accurate and relevant recommendations. Kumar et al. (2015) demonstrated that their approach outperformed traditional collaborative filtering methods, especially in cases where users had sparse interaction data in one of the domains [5]. Fernández-Tobías et al. (2019) focused on addressing the user cold-start problem in cross-domain recommendation systems. They proposed a method that incorporates item metadata into matrix factorization models to mitigate the sparsity issue often encountered when new users interact with the system. Their approach combines collaborative filtering with content-based features, leveraging item attributes to enhance the personalization of recommendations. The results showed that integrating item metadata significantly improved the recommendation performance, especially for users with limited interaction history. This method is particularly useful in domains where the availability of user-item interaction data is scarce, such as in the case of newly launched products or services [6]. Loni et al. (2014) applied factorization machines to cross-domain collaborative filtering, which can model higher-order interactions between users and items across multiple domains. Their work demonstrated that factorization machines were particularly effective in modeling interactions in multi-domain environments. By capturing complex relationships, factorization machines were able to provide more accurate recommendations, especially in cases where traditional methods struggled with sparse data. The study concluded that factorization machines could serve as a powerful tool for improving the performance of cross-domain recommendation systems, making them highly suitable for applications where data from diverse sources need to be integrated [7]. Yu et al. (2021) proposed a mixed heterogeneous factorization model to handle non-overlapping domains in cross-domain recommendation systems. This approach is designed to tackle the challenges posed by domains with very little overlap in user behavior and item preferences. The mixed heterogeneous factorization model successfully integrates information from multiple domains, even when there is limited shared data. Through extensive experiments, the authors showed that their model outperformed other cross-domain methods in terms of recommendation accuracy and robustness, especially in scenarios where domains were largely independent of one another. This method offers a promising solution for recommending products or services from different domains that do not share significant overlap in terms of user

interests [8]. Wang et al. (2021) introduced a cross-domain recommendation method that incorporates user personality traits into matrix factorization models. Their research found that user personality plays a significant role in shaping preferences across different domains. By incorporating personality factors into the recommendation process, they were able to enhance the personalization of recommendations, particularly in domains where user preferences were highly variable. Their study demonstrated that personality-aware models lead to more diverse and personalized recommendations, which is particularly beneficial for applications in areas like entertainment and social media, where user preferences are diverse and evolving [9]. Khan et al. (2017) conducted a systematic literature review on cross-domain recommender systems, focusing on the various approaches and techniques used to address the challenges of domain heterogeneity and cold-start problems. They reviewed a wide range of methods, from matrix factorization and tensor decomposition to hybrid models that combine collaborative and content-based filtering. Their work identified several key trends in the field, including the increasing use of semantic similarity and factorization techniques to bridge domain gaps. The review provides valuable insights into the current state of cross-domain recommendation systems and highlights potential directions for future research, including the integration of deep learning and advanced factorization methods to improve recommendation accuracy [10]. These studies collectively highlight the importance of leveraging advanced factorization techniques, semantic similarity, and user modeling to improve the accuracy and effectiveness of cross-domain recommendation systems. By combining multiple domains and incorporating auxiliary information such as metadata, user personality, and item attributes, these approaches address many of the challenges traditionally faced by recommendation systems, such as data sparsity and the cold-start problem. The findings from these studies provide a strong foundation for developing more effective and personalized recommendation systems across diverse applications.

III. Objectives of the Study

The objectives of this study are as follows:

1) **To explore the effectiveness of cross-domain recommendation systems:** This study aims to investigate the use of cross-domain recommendation systems, specifically focusing on how data from multiple domains can be utilized to improve recommendation accuracy. The research will examine the integration of auxiliary information such as metadata, user preferences, and item attributes to enhance personalization. .
2) **To analyze the role of matrix and tensor factorization techniques:** The study will delve into the effectiveness of matrix and tensor factorization methods in cross-domain settings. The research will evaluate these methods in terms of their ability to address challenges like data sparsity, cold-start problems, and domain heterogeneity, which often affect the performance of traditional recommendation systems.
3) **To compare various cross-domain recommendation approaches:** The study will compare different techniques used in the cross-domain recommendation, such as multidimensional tensor factorization, hybrid models, and semantic similarity approaches. The goal is to assess the strengths and weaknesses of each method in various application scenarios.
4) **To improve recommendation performance by incorporating external knowledge:** This study aims to examine how external knowledge sources, such as semantic embeddings, ontologies, and user personality traits, can be integrated into cross-domain recommendation systems. The research will evaluate how incorporating these additional features improves the personalization and relevance of recommendations.
5) **To propose a novel cross-domain recommendation model:** Based on the insights gained from the above objectives, this study aims to propose a new, hybrid recommendation model that can effectively combine data from multiple domains and leverage both collaborative filtering and content-based methods. The model will be tested for performance improvements in terms of accuracy, diversity, and robustness.

IV. Data Collection

We have used two datasets, i.e., booksdata.csv and reviews.csv. The datasets used for this study were obtained from Kaggle and can be accessed via the following link: EDA for Amazon Books Reviews. This dataset is a part of the Amazon review dataset, which contains product reviews and metadata spanning from May 1996 to July 2014. The dataset provides feedback from 3 million records of users on 212,404 unique books with 10 features, allowing for an extensive analysis of Amazon book reviews.Products Reviews dataset consists of 568,454 rows and 10 features that consist of 212,404 unique book titles.

The dataset consists of two main files:

1. Books Ratings Dataset

The Reviews file contains feedback from 3 million records of users on 212,404 unique books. The table below summarizes the key attributes in the Books Ratings file:

Feature	Description
`id`	The ID of the book
`Title`	The title of the book
`Price`	The price of the book
`User_id`	The ID of the user who rates the book
`profileName`	The name of the user who rates the book
`review/helpfulness`	Helpfulness rating of the review (e.g., 2/3)
`review/score`	Rating from 0 to 5 for the book
`review/time`	Time of the review submission
`review/summary`	The summary of the text review
`review/text`	The full text of the review

TABLE I
Attributes of the Books Rating Dataset

2. Products Dataset

The Products Details file contains details and contains 568,454 records with 10 features. The table below summarizes the key attributes in the product details file:

Feature	Description
`Id`	Unique identifier for each review
`ProductId`	Identifier for the product being reviewed
`UserId`	Identifier for the user who submitted the review
`ProfileName`	Profile name of the user
`HelpfulnessNumerator`	Number of users who found the review helpful
`HelpfulnessDenominator`	Total number of users who voted on the review's helpfulness
`Score`	Rating score given by the user (1–5)
`Time`	Timestamp of the review
`Summary`	Short summary of the review
`Text`	Full text of the review

TABLE II
ATTRIBUTES OF THE PRODUCT REVIEWS DATASET

This datasets provides valuable information for performing exploratory data analysis (EDA) on Amazon book ratings and product ratings, helping to gain insights into user preferences, ratings, and book attributes. It can be used for analyzing sentiment, review trends, and other patterns within the domain of book recommendations.

For more information, you can access the datasets via the following link: EDA for Amazon Books Reviews.Products Reviews dataset

V. EXPLORATORY DATA ANALYSIS (EDA)

A. Dataset Overview

The analysis begins by reviewing the key features of the two datasets: the Product Reviews and Book Reviews datasets. These datasets consist of reviews and ratings for products and books, respectively. The products dataset has 568454 records with 10 features, while the books dataset has 3000000 records, i.e., 3 million with 10 features.

B. Missing Values and Data Cleaning

Before beginning the analysis, we check both datasets for missing values.

1) Product Reviews: The following columns have missing values:

- `ProfileName`
- `Summary`

These variables don't make any difference for our analysis, as we won't consider those variables for analysis, so we have left it as it is.

2) Book Reviews: The following column has missing values:

- `Title`
- `Price`
- `Userid`
- `profileName`
- `review/summary`
- `review/text`

Missing values for these variables are also not considered for further cleaning, as we only consider data for matched users in both datasets.

C. Renaming Columns for Consistency

The columns in both datasets are renamed for consistency and ease of analysis:

- For the **Product Reviews Dataset**, columns such as `ProductId` are renamed to `Product_ID`.
- For the **Book Reviews Dataset**, columns such as `User_id` and `review/score` are renamed to `User_ID` and `Rating`.

D. Rating Distribution Visualization

We visualize the distribution of ratings for both datasets to understand user feedback trends.

1) Product Reviews Rating Distribution: The ratings range from 1 to 5, and the distribution shows how many reviews correspond to each rating. A histogram or bar chart is plotted to visualize this distribution.

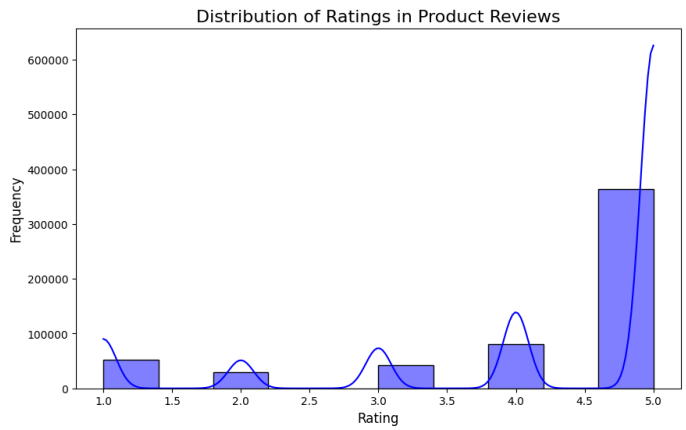

Fig. 1. Distribution of Ratings in Product Reviews Dataset

2) Book Reviews Rating Distribution: Similarly, the ratings for the Book Reviews dataset are visualized, showing the frequency of each rating.

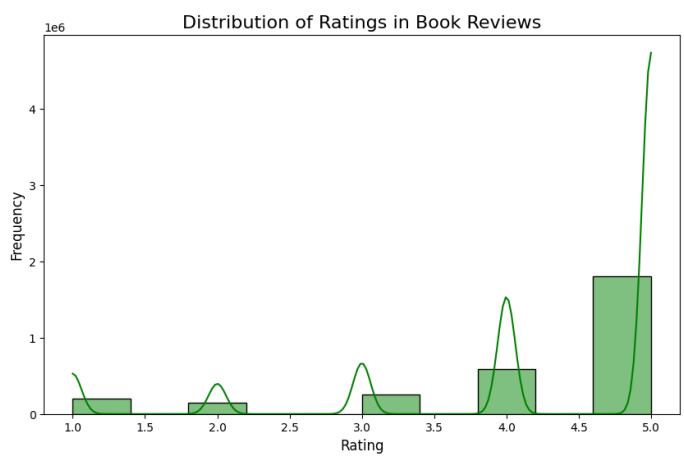

Fig. 2. Distribution of Ratings in Book Reviews Dataset

E. Sampling the Data

Given the size of both datasets, we sample a small portion to make analysis more manageable.

1) Product Reviews Dataset Sample: A 0.05% sample is taken from the Product Reviews dataset, which results in:

$$\text{Product Reviews Sample Size: } 28,423 \text{ rows}, 10 \text{ columns}$$

2) Book Reviews Dataset Sample: A 0.005% sample is taken from the Book Reviews dataset, which results in:

$$\text{Book Reviews Sample Size: } 15,000 \text{ rows}, 10 \text{ columns}$$

F. Merging Both the Datasets

Initially, we need to merge datasets with respect to userID and product ID by pivoting to find the reviews of the same users for different products.

Identifying Common Users

To ensure meaningful analysis, we identify users who have interacted with both products and books. This allows us to explore cross-domain preferences.

Filtering the Datasets for Common Users

We filtered the datasets to include reviews only from these common users, and then we validated the number of unique users in the filtered datasets.

Averaging Ratings for Each User-Item Pair

To minimize noise and maintain consistency, we group ratings by user-item pairs and calculate the average rating per pair.

VI. Data Analytics

A. User-Item Matrix Construction

Next, we built a user-item matrix for both datasets, with users as rows and products or books as columns. Ratings are used to populate the matrix.

1) Product Reviews User-Item Matrix: The matrix is constructed by mapping each user to the products they reviewed, with the ratings as the values. Missing values are filled with zeros.

2) Book Reviews User-Item Matrix: The same process is applied to the Book Reviews dataset.

B. Matrix Factorization

We use Singular Value Decomposition (SVD) for matrix factorization to reduce dimensionality and uncover latent factors.

1) Product Reviews Matrix Factorization: SVD is applied to the Product Reviews user-item matrix to predict missing ratings and reduce dimensionality. The reconstructed matrix provides predicted ratings.

2) Book Reviews Matrix Factorization: SVD is similarly applied to the Book Reviews dataset, with the reconstructed matrix showing predicted ratings.

VII. Data Visualization and Results Reports

The following visualizations provide an overview of the recommendations:

Books Recommended Based on Product Preferences

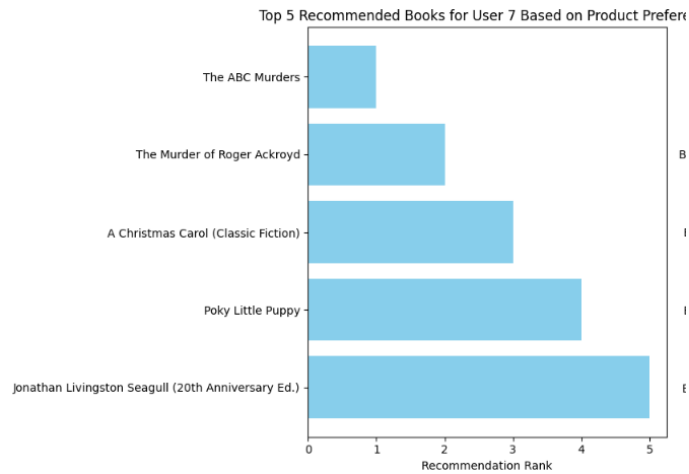

Figure 1: Top 5 Books Recommended for User 7 Based on Product Preferences.

Products Recommended Based on Book Preferences

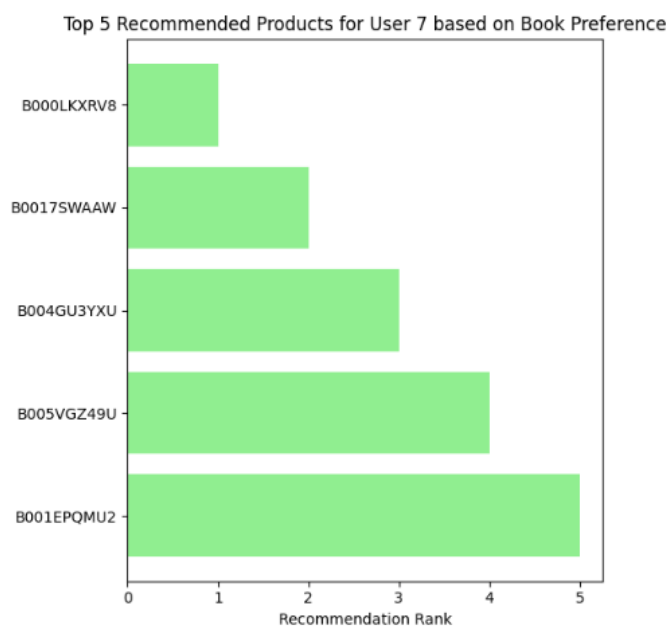

Figure 2: Top 5 Products Recommended for User 7 Based on Book Preferences.

A. Results

The top recommendations for User 7 based on their preferences are as follows:

VIII. Hypothesis

The hypothesis for this study is formulated as follows:

Rank	Books (Based on Product Preferences)
1	The ABC Murders
2	The Murder of Roger Ackroyd
3	A Christmas Carol (Classic Fiction)
4	Poky Little Puppy
5	Jonathan Livingston Seagull (20th Anniversary Ed.)

TABLE III
TOP 5 BOOKS RECOMMENDED FOR USER 7 BASED ON PRODUCT PREFERENCES

Rank	Products (Based on Book Preferences)
1	B000LKXRV8
2	B0017SWAAW
3	B004GU3YXU
4	B005VGZ49U
5	B001EPQMU2

TABLE IV
TOP 5 PRODUCTS RECOMMENDED FOR USER 7 BASED ON BOOK PREFERENCES

A. Hypothesis

The integration of user preferences across different domains (e.g., product reviews and book reviews) can provide meaningful and personalized cross-domain recommendations that are both relevant and effective in enhancing user satisfaction.

B. Null Hypothesis (H_0)

User preferences in one domain (e.g., product reviews) do not have a significant impact on the relevance of recommendations in another domain (e.g., book reviews).

C. Alternative Hypothesis (H_1)

User preferences in one domain (e.g., product reviews) significantly influence and improve the relevance of recommendations in another domain (e.g., book reviews).

This hypothesis provides the basis for evaluating the effectiveness of cross-domain recommendation systems by analyzing the relationship between user preferences across multiple domains. Each model was trained on the dataset, with cross-validation used to assess the model's performance. The loss function used was categorical cross-entropy, and accuracy was the primary evaluation metric.

IX. CONCLUSION AND FUTURE WORK

This study demonstrates the potential of cross-domain recommendation systems by leveraging user preferences in one domain (e.g., product reviews) to generate relevant and personalized recommendations in another domain (e.g., book reviews). The results highlight the effectiveness of integrating Singular Value Decomposition (SVD) for dimensionality reduction and collaborative filtering techniques to uncover latent relationships between users and items across domains. The findings suggest that such approaches can enhance user satisfaction by providing diverse and meaningful recommendations. For future work, we propose extending this framework to include additional domains, such as book or product domains, to evaluate its generalizability and scalability. Moreover, incorporating advanced neural network models, such as attention mechanisms and graph-based embeddings, can improve the precision of recommendations by capturing complex user-item interactions. Finally, integrating explicit user feedback into the system can refine recommendation accuracy and address potential biases in the data.

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